Intermediate deliverable:

Data Preparation and ETL Methods

High-Five

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Data sources

Analysis 1 & 2

Dataset name	Source	Link	Coverage	Key variables	Format	Access method
Seoul Digital Competency Survey	Seoul AI Foundation	https://saif.or.kr/dataroom/2564	Geographic: Seoul Time: 2023 used	devices, skills, safety; survey weights	CSV/XLSX	data provision upon official request
Seoul Survey	Seoul Open Data	https://data.seoul.go.kr/dataList/OA-15564/F/1/datasetView.do	Geographic: Seoul Time: 2023, 2024	digital service use; demographics, income, housing, district/year		Public Open Data

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Data sources

About Seoul Digital Competency Survey

---------- Forwarded message ---------
보낸사람: **박선미** <triton@saif.or.kr>
Date: 2026년 2월 11일 (수) AM 9:25
Subject: Re:서울시민 디지털역량 실태조사 원시데이터 자료이용 신청
To: 양재혁 <jaehyeok7243@gmail.com>

안녕하세요.

요청하신 '2021/2023년 서울시민 디지털역량실태조사' 원자료 송부드립니다.

첨부된 자료로 공동연구(대표 1인 자료신청) 진행하시면 되고, 자료이용 준수 서약서 확인하시어 이용 부탁드립니다.

감사합니다.

박선미 드림.

SAIF 서울AI재단 Seoul AI Foundation
AI데이터분석팀
박선미 팀장 / 경영학(MIS) 박사
02-570-4680 / 010-4220-8810 / triton@saif.or.kr
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보내는사람 : 양재혁 <jaehyeok7243@gmail.com>

받는사람 : '박선미'<triton@saif.or.kr>

보낸 날짜 : 2026-02-10 18:19:26

제목 : 서울시민 디지털역량 실태조사 원시데이터 자료이용 신청

서울시민 디지털역량 실태조사 원시데이터 자료이용 신청합니다.
추가적으로 혹시 공동 연구 진행할 경우 연구진 전원이 자료이용신청해야되는지 궁금합니다.

첨부파일 4개 • Gmail에서 검사함 ⓘ   Drive에 모두 추가

raw data(공개용).zip
코드북_2021_서울시...
코드북_2023 서울시...
코드북_2023_서울시...

Requested raw data separately on Feb 11, 2026
Obtained via official channels

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ITU

Data sources

Analysis 1 & 2

Dataset name	Source	Link	Coverage	Key variables	Format	Access method
Cellular Base Station Data	Spectrum Resource Mgmt System	https://spectrummap.kr/gis/mobile_service.do?menuNo=300480	Geographic: Seoul Time: 2023, 2024	4G/5G cell-site locations	CSV/XLSX	Public Open Data
NIA Communication Quality (CQ)	Korean Public Data Portal	https://www.data.go.kr/data/15120591/fileData.do	Geographic: Seoul Time: 2023, 2024	district-level download speed		
Public Free Wi-Fi Installation Data	Local Admin WiFi License Data	https://www.localdata.go.kr/lif/lifeCtacDataView.do	Geographic: Seoul Time: 2023, 2024	public Wi-Fi installation locations		
Telecommunication Indicators	SKT Telecom Data	https://www.data.go.kr/data/15127168/fileData.do#	Geographic: Seoul Time: 2023, 2024	mobile data use; online service days; fee delinquency rate		
Average Monthly Household Income by District	Seoul Commercial Analysis Service	https://data.seoul.go.kr/dataList/OA-22167/S/1/datasetView.do	Geographic: Seoul Time: 2023, 2024	average monthly household income		

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Data sources

Analysis 3

Dataset name	Source	Link	Coverage	Key variables	Format	Access method
Daangn Community Post	Daangn Market	daangn.com/kr/community/s	Geographic: Seoul dong fields Time: 2024-2026 crawl period	post text; gu/location; createdAt; subject/category	CSV/XLSX	Web Scraping
Seoul Living/Floating Population	Seoul Open Data	data.seoul.go.kr OA-14991	Geographic: Seoul Time: 2023-2026 used as available	hourly/dong population; daytime population; age shares; dong/district code		Public Open Data

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Data Preparation Methods

Analysis 1: UMC Index

Connectivity: turning network infrastructure into comparable district measures

- Unit of comparison: each Seoul district in each study year.
- Data used: Cellular Base Station Data locations, measured download speed, district boundaries, and daytime population.

Preparation steps

- Count each physical base-station site only once.
- Match all records to the same Seoul district code system.
- Convert station counts into population-adjusted density measures.
- Average measured download speed at the district-year level.
- Place all connectivity indicators on a common scale before combining them.

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Data Preparation Methods

Analysis 1: UMC Index

Availability for Use: measuring whether digital services are practically usable

- Unit of comparison: each Seoul district in each study year.
- Data used: telecom usage(SKT Telecom Data), online service days, public Wi-Fi installations, population denominators, and district codes.

Preparation steps

- Aggregate neighborhood-level telecom data to district level using daytime floating population weights.
- Convert public Wi-Fi installations into resident-adjusted density.
- Usage, service-day, and Wi-Fi indicators on a common scale before combining them.

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Data Preparation Methods

Analysis 1: UMC Index

Affordability: separating ability to pay from payment burden

- Unit of comparison: each Seoul district in each study year.
- Data used: district household income and telecom fee delinquency.

Preparation steps

- Use income as the positive resource side of affordability.
- Use telecom fee delinquency as the burden side of affordability.
- Aggregate delinquency from neighborhood to district level with population weights.
- Reverse the burden indicator so that higher final values consistently mean better conditions.
- Combine the resource and burden sides after standardization.

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Data Preparation Methods

Analysis 1: UMC Index

Devices, Skills, and Safety: converting survey answers into district-level conditions

- Data used: Seoul Digital Competency Survey with official survey weights.

Constructed dimensions

- Devices: whether residents have and use core digital devices.
- Digital skills: task-based ability items grouped into a skills score.
- Safety and security: safe-use behavior and security-awareness items.

Preparation steps

- Remove invalid and nonresponse codes before scoring.
- Apply survey weights so district averages represent the sampled population.
- Use the available survey wave as a time-invariant district input for both study years.

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Data Preparation Methods

Analysis 2: Multilevel Survey Dataset

Goal: connect individual digital-use behavior with district-level UMC context.

Survey pooling

- Combine the two Seoul Survey years into one respondent-level analysis frame.
- Link each respondent to the correct district-year UMC context using district and year keys.

Outcome construction

- Convert digital service-use responses to a common score range and sum them into one digital-use measure.
- Assign zero use only when the respondent is identified as a non-user.

Variable harmonization

- Standardize field names that changed between survey years before pooling.
- Create demographic and household controls needed for the multilevel model.
- Keep selected binary outcomes for robustness checks.

Scaling and centering

- Place district-level predictors on comparable scales; reverse cost-burden indicators.
- Center age and income using constants saved with the processed dataset.
- Document every transformation in the analysis codebook.

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Data Preparation Methods

Analysis 3: Daangn Community Posts

Goal: translate local community text into district-level UMC evidence.

Corpus cleaning

- Remove deleted, blocked, and empty-text posts before analysis.
- Standardize text, time, category, and location fields when available.

Candidate screening

- Use UMC keyword dictionaries to identify potentially relevant posts.
- Keep all matched UMC dimensions at the screening stage instead of forcing one label.

Scoring and aggregation

- Score selected posts for UMC relevance, dimension, confidence, and problem type.
- Aggregate scored evidence by district and dimension for Bayesian updating.

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Missing Data Rules

Analysis 1: UMC Index

Principle: clean each source before constructing composite scores.

Indicator-level rules

- Convert blanks and nonresponse codes to missing before calculation.
- Use official population denominators for density and per-capita measures when available.
- Keep fallback denominator logic only for reproducibility, not as the main rule.

Survey-derived dimensions

- Exclude invalid survey responses before constructing devices, skills, and safety measures.
- Use survey weights only when both response and weight are valid.

Composite construction

- Apply the same normalization and dimension formula after source-specific cleaning.

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Missing Data Rules

Analysis 2: Survey Dataset

Principle: use rule-based handling that matches the survey meaning.

Standardization

- Read blank and null-like survey entries as missing values before pooling.
- Harmonize variables that changed names or coding between survey years.

Rule-based handling

- Assign zero digital use only to respondents identified as non-users.
- Apply the documented income nonresponse rule before centering.

Modeling file

- Remove unresolved missing values in the dependent variable or required model controls.
- Leave robustness-only gaps in robustness variables rather than imputing them into the main model.

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Missing Data Rules

Analysis 3: Daangn Community Posts

Principle: keep uncertainty visible instead of filling text-derived gaps artificially.

Invalid-resource filtering

- Remove unusable posts before keyword screening.
- Keep posts with usable text and available location information for downstream matching.

Location context gaps

- Do not impute unmatched official-dong or living-population context.
- Use district labels for aggregation when finer neighborhood context is unavailable.

Uncertainty handling

- Keep unassigned dimensions as zero-probability fields.
- Use Bayesian aggregation so sparse text evidence is represented with uncertainty.

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Data Management

Analysis 1: UMC Index

Goal: make every index value traceable back to its source and transformation rule.

Storage structure

- Separate raw sources, cleaned indicators, normalized dimensions, and final index tables.
- Store each study year separately to support year-by-year review.

Audit trail

- Keep source-specific files before joining so transformations remain traceable.
- Store codebook, descriptives, and summaries beside the processed outputs.

Documentation

- Record source role, aggregation rule, scaling direction, and dimension formula in the codebook.

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Data Management

Analysis 2: Multilevel Survey Dataset

Goal: preserve a clean path from raw survey responses to the modeling file.

Storage structure

- Save the processed model frame in both efficient and inspection-friendly formats.
- Store centering constants and panel metadata separately from the model frame.

Audit trail

- Keep raw survey files separate from processed analysis files.
- Maintain district-year UMC predictors as external contextual inputs before merging.

Documentation

- Use the analysis codebook to document variable definitions, transformations, scales, and source roles.

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Data Management

Analysis 3: Daangn Community Posts

Goal: keep the text-processing and Bayesian-model stages auditable.

Storage structure

- Save cleaned posts, screened candidates, scored posts, aggregates, posterior tables, and figures separately.
- Keep scoring outputs separate from Bayesian posterior outputs.

Audit trail

- Preserve raw, candidate, and scored stages so each post-level decision can be traced.
- Use district-dimension aggregates as the bridge from text scoring to posterior estimation.

Documentation

- Version posterior outputs by model setting so sensitivity checks remain transparent.

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ETL Workflow

Analysis 1: UMC Index

EXTRACT

- Collect infrastructure, service-use, affordability, competency, and population-context sources.

TRANSFORM

- Align district codes and remove duplicate infrastructure records.
- Convert counts and rates into comparable district-year indicators.
- Use population weights where neighborhood data are aggregated upward.
- Standardize indicators and combine them into the six UMC dimensions.

LOAD

- Save versioned UMC tables and documentation for downstream analyses.

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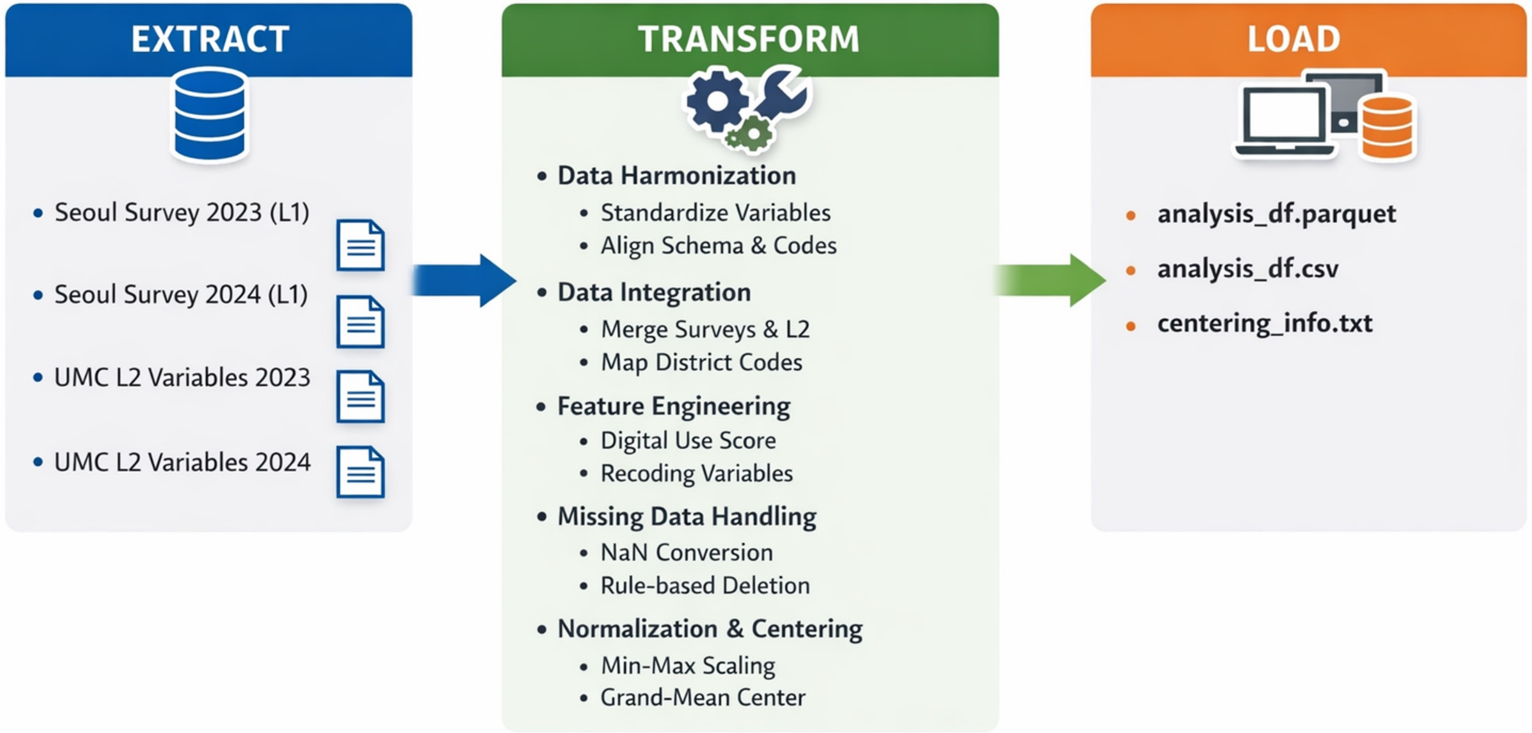
ETL Workflow

Analysis 2



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ETL Workflow

Analysis 3: Daangn Community Posts

EXTRACT

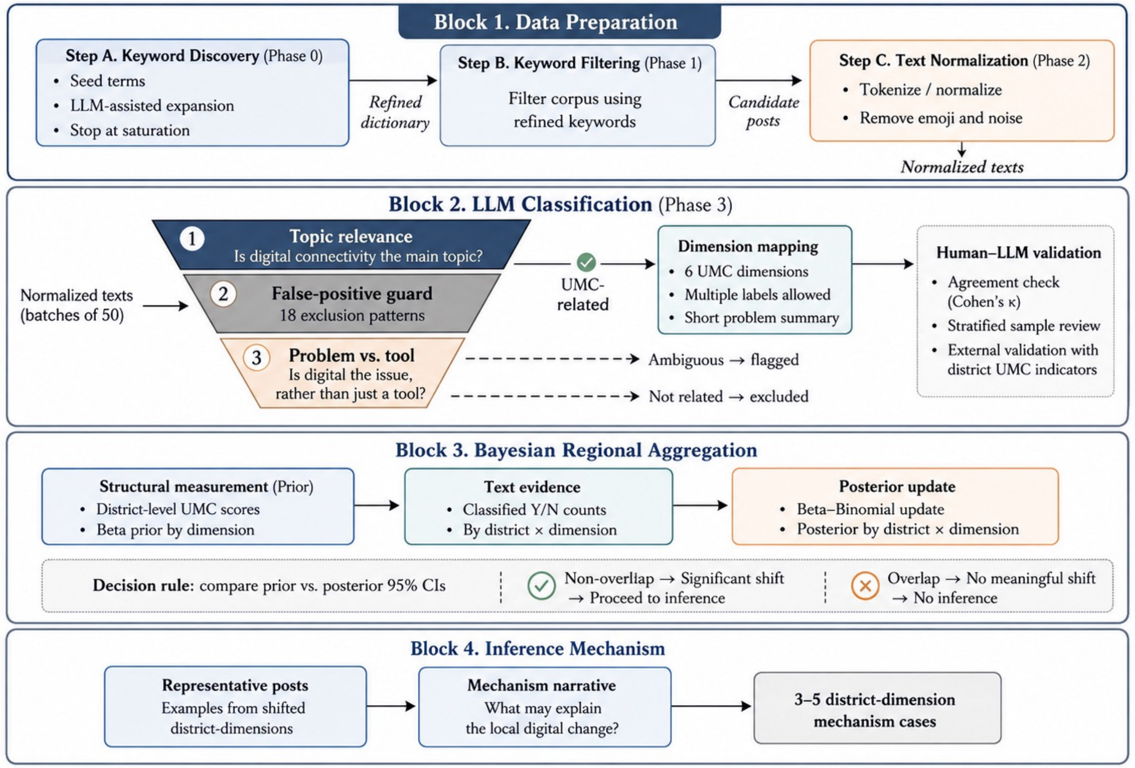
- Collect clean community posts, district UMC priors, neighborhood mapping, and population context.

TRANSFORM

- Screen posts with UMC keywords and score selected posts with UMC probabilities.
- Map neighborhood context when available and aggregate labels to district-dimension counts.
- Update measured UMC priors with text evidence using the Bayesian model setting.

LOAD

- Save scored-post data, summary tables, posterior tables, and figures.



Posts are treated as likelihood evidence rather than direct measures of regional UMC.
Inference is made only when posterior shifts are significant and within causal bounds.

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Others

Team Notion Page: https://www.notion.so/ITU-Data-Hackathon-2e9594c68227801ca869dab7e2428f3f?source=copy_link

Github: Will address this soon

Report_draft: Attatched File

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